

mophat524 / Aviation

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

0 stars

0 forks

0 watching

Branches

Activity

Tags

Public repository

master had recent pushes 8 minutes ago

Compare & pull request

2 Branches

0 Tags

Go to file

Go to file

Add file

Code

This branch is 8 commits ahead of, 1 commit behind main .

Contribute

	mophat524 added aviation presentation	aeb86ec · 8 minutes ago	
	Aviation Dashboard.pdf	added Tableau visuals	1 hour ago
	Aviation.ipynb	first commit	2 hours ago
	AviationData.csv	first commit	2 hours ago
	AviationNotebook.pdf	added aviation presentation	11 minutes ago
	Aviation_Project_Presentatio...	added aviation presentation pdf	41 minutes ago
	Cleaned_AviationData.csv	first commit	2 hours ago
	README.md	Add README for Aviation Accident ...	2 hours ago
	TableauAviationAnalysis.twb	first commit	2 hours ago
	~TableauAviationAnalysis_3...	first commit	2 hours ago

README



Aviation Accident Analysis Project



Overview

This project analyzes aviation accident data to uncover patterns, trends, and insights related to aircraft incidents across various countries and years.

The dataset contains detailed information about each aviation event — including its location, date, severity, weather conditions, and aircraft details.

The goal is to use this data to understand **factors influencing accident severity**, identify **risk patterns**, identify **operational risk**, identify **market risk** and support **data-driven safety recommendations** for the aviation industry.

Problem Statement

Aviation safety is critical, yet accidents still occur due to multiple complex factors such as weather, human error, or technical malfunction.

This project aims to answer key analytical questions, including:

- What are the most common causes of aviation accidents?
- Which aircraft makes and models are most frequently involved in accidents?
- How have aviation accident trends evolved over the years?

By performing exploratory data analysis (EDA) and visualization, this project seeks to **improve situational awareness**

Dataset Description

The dataset contains detailed records of aviation events.

Below are the key columns and their meanings:

Column	Description
EventId	Unique identifier for each event
InvestigationType	Indicates if it was an accident or incident
AccidentNumber	Official accident report number
EventDate	Date of the event
Location	City/region where the event occurred
Country	Country of occurrence
Latitude	Geographic coordinate (north-south position)
Longitude	Geographic coordinate (east-west position)
InjurySeverity	Severity of injuries (e.g., Fatal, Serious, Minor, None)
AircraftDamage	Extent of aircraft damage (e.g., Destroyed, Substantial)
AircraftCategory	Type/category of aircraft (e.g., Airplane, Helicopter)
Make	Manufacturer of the aircraft
Model	Specific model of the aircraft
PurposeofFlight	Purpose (e.g., Personal, Commercial, Training)
TotalFatalInjuries	Number of fatalities
TotalSeriousInjuries	Number of serious injuries
TotalMinorInjuries	Number of minor injuries

Column	Description
TotalUninjured	Number of uninjured occupants
WeatherCondition	Weather at the time (e.g., VMC, IMC)
ReportStatus	Status of investigation (e.g., Final, Preliminary)
Year	Year of occurrence



Methodology

1. Data Cleaning

- Handle missing values
- Standardize column names and data types
- Convert dates to proper datetime format

2. Exploratory Data Analysis (EDA)

- Frequency of accidents by year, country, and aircraft make
- Visualization of accident locations using coordinates

3. Data Visualization

- Bar charts for severity distribution
- Geospatial scatter plots (using Latitude & Longitude)

4. Insights & Reporting

- Identify top risk factors
- Use key performance indicators to track injuries
- Use maps to show market risks in other countries
- Use time series to show injuries that have occurred in years
- Summarize findings for safety recommendations



Technologies Used

- Python
 - pandas → Data cleaning and manipulation
 - matplotlib & seaborn → Data visualization
 - plotly / folium → Interactive geographic mapping
- Visual Studio Code Notebook for analysis and documentation
- Git & GitHub for version control



Example Questions to Explore

- Which year had the highest number of aviation accidents?
- Which countries report the highest number of accidents?

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)